



Louis de Broglie

德布罗意物质波

"The universe is not only stranger than we imagine, it is stranger than we can imagine."

—Louis de Broglie

Zephyr



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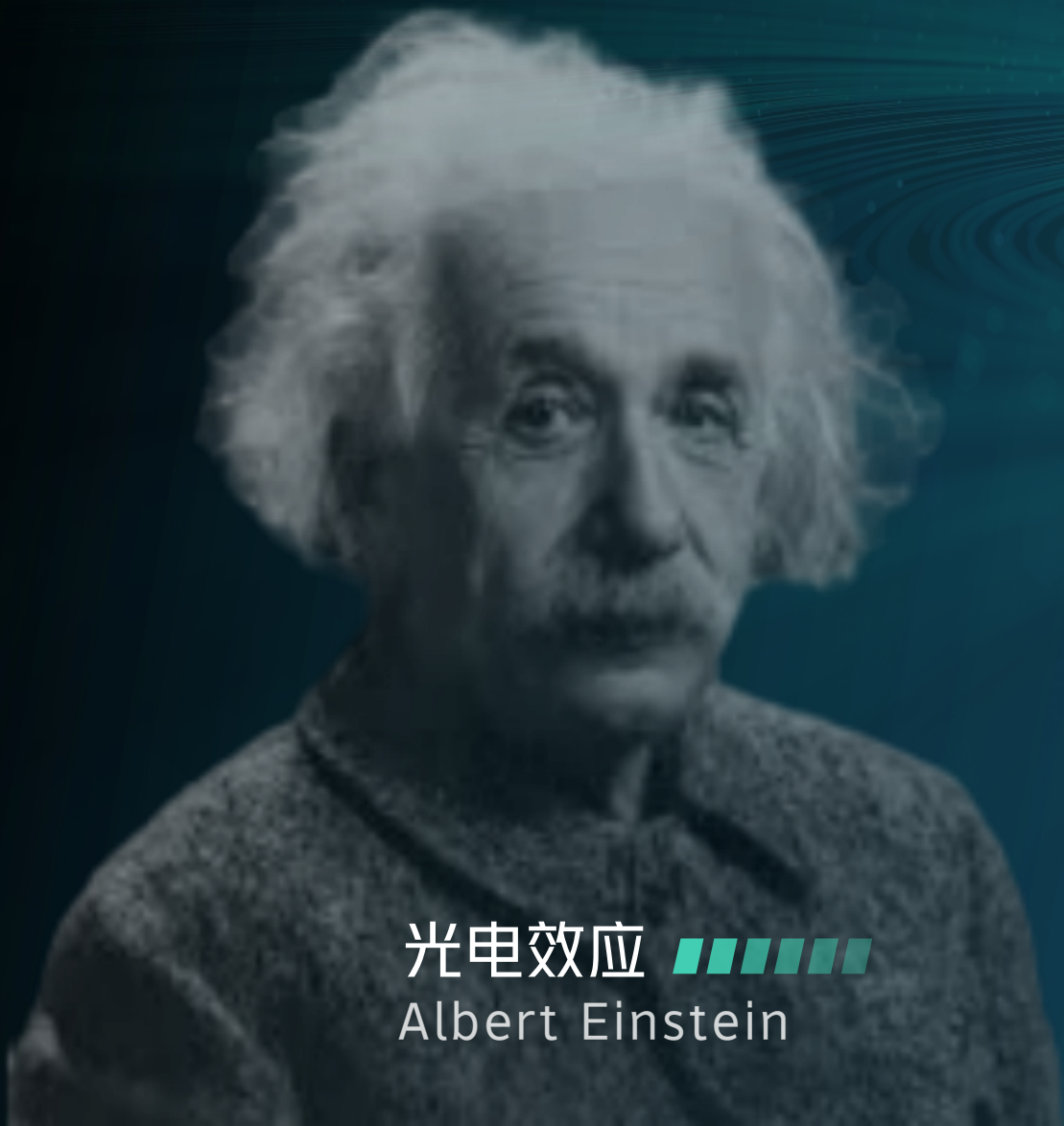


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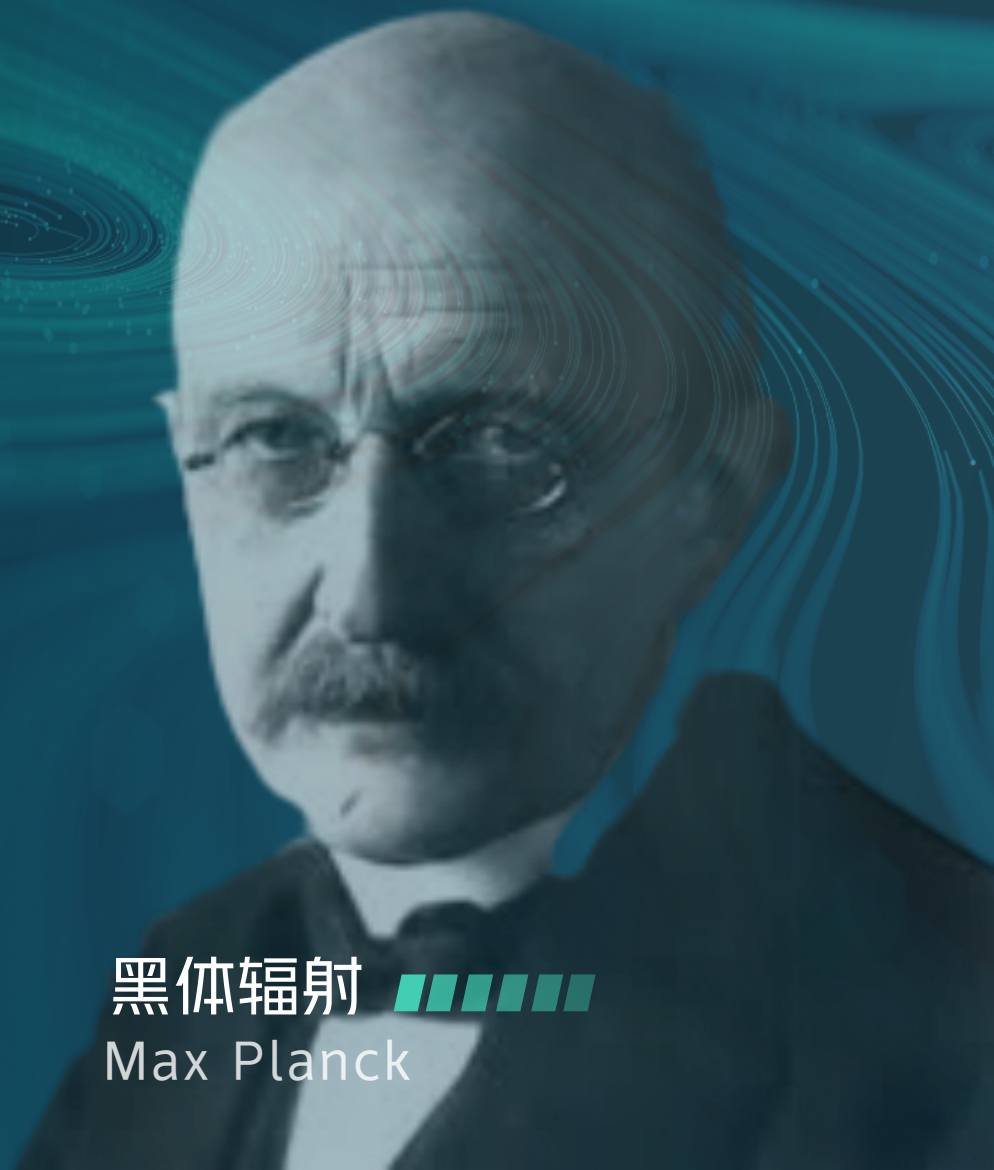
波粒二象性

黑体辐射&光电效应

01 光电效应&黑体辐射



光电效应 
Albert Einstein



黑体辐射 
Max Planck



02.

德布罗意物质波假说

Davisson-Germer experiment

02 Definition

de Broglie relation

a particle's momentum p is inversely proportional to its wavelength λ

de Broglie wavelength

the wavelength, λ , associated with a particle with momentum p through the Planck constant, h

wavelength associated with a particle that is moving.

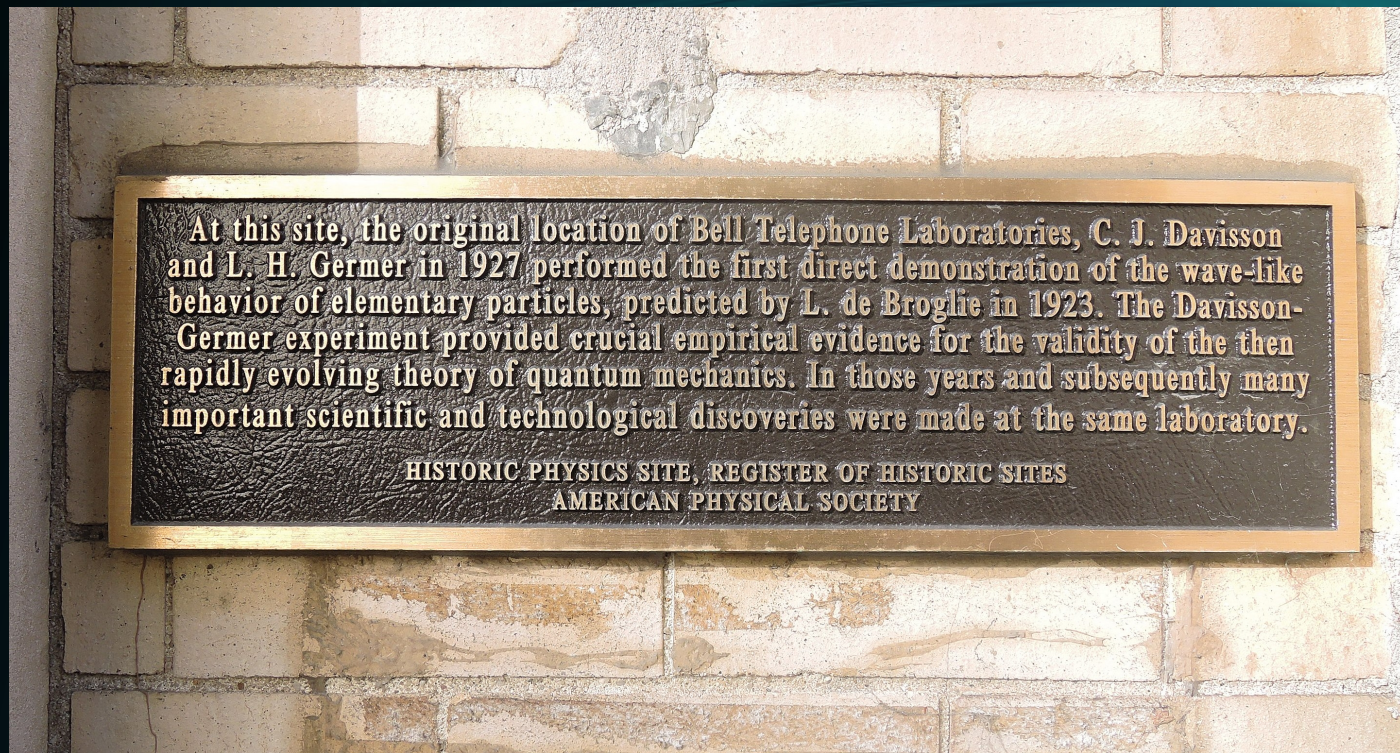
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戴维森-革末实验

Davisson-Germer experiment



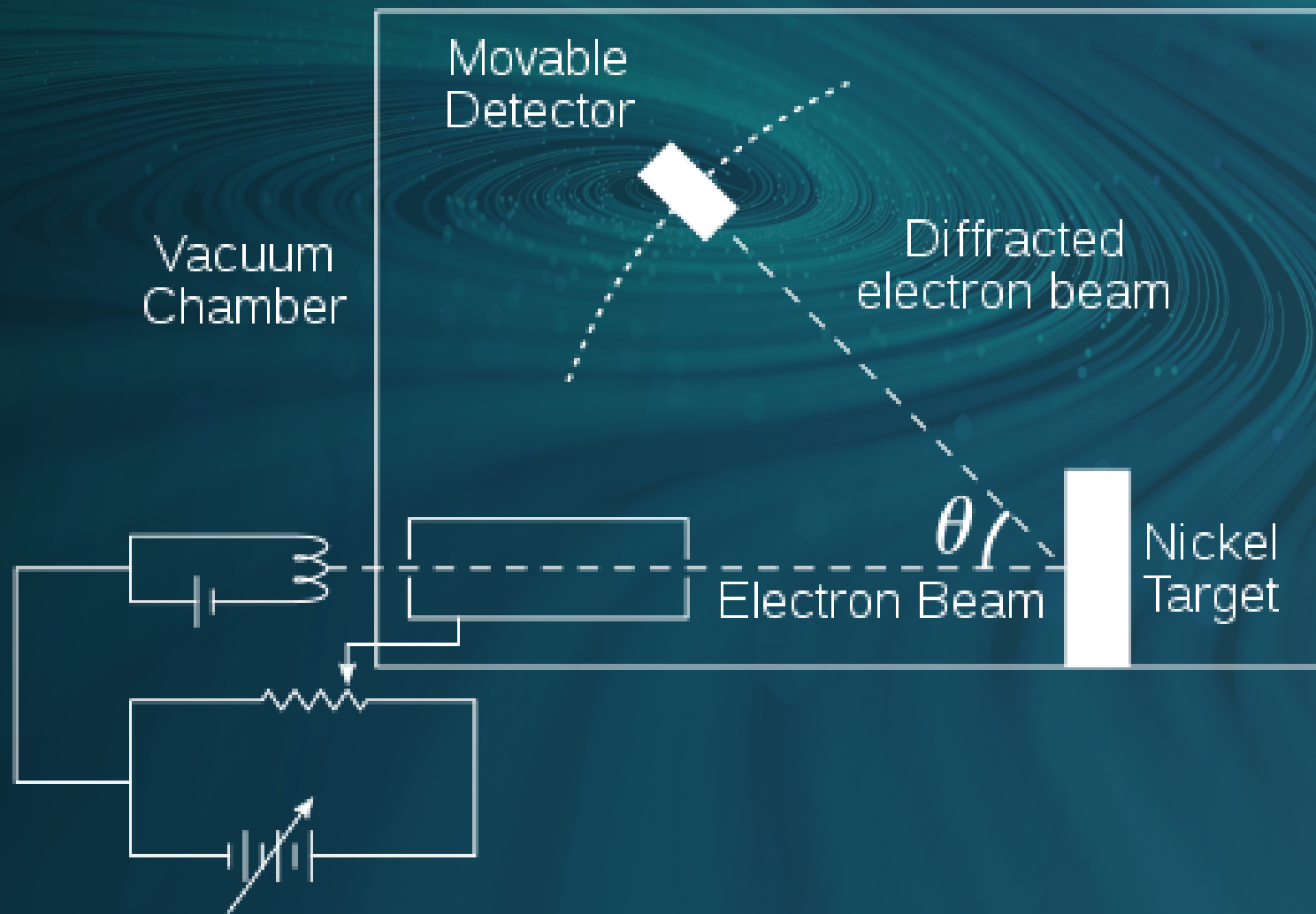
02 戴维森-革末实验



曼哈顿的美国物理学会
纪念牌匾纪念该实验

02 戴维森-革末实验

实验设置



02 戴维森-革末实验

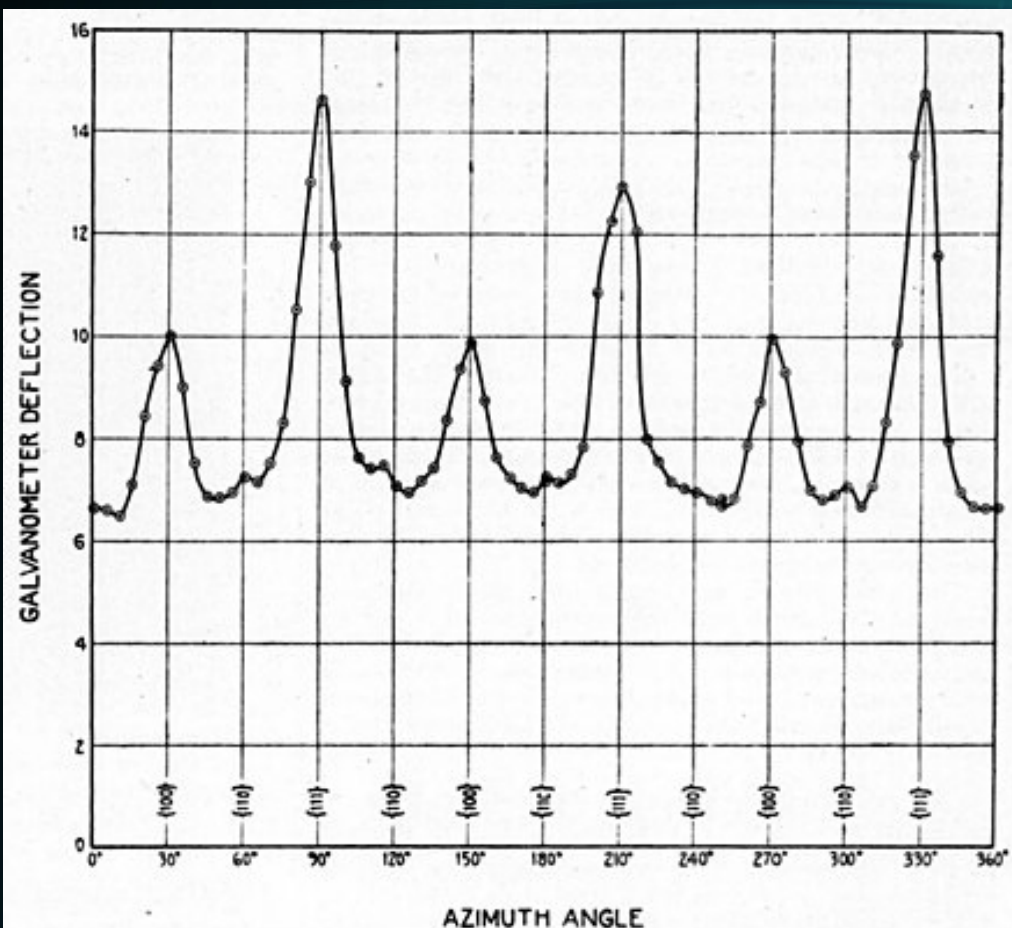


FIG. 2.—Intensity of electron scattering vs. azimuth angle—54 volts, co-latitude 50°.

电流与电子束方位角关系图。峰和谷的存在与衍射图一致，表明电子具有波状性质。



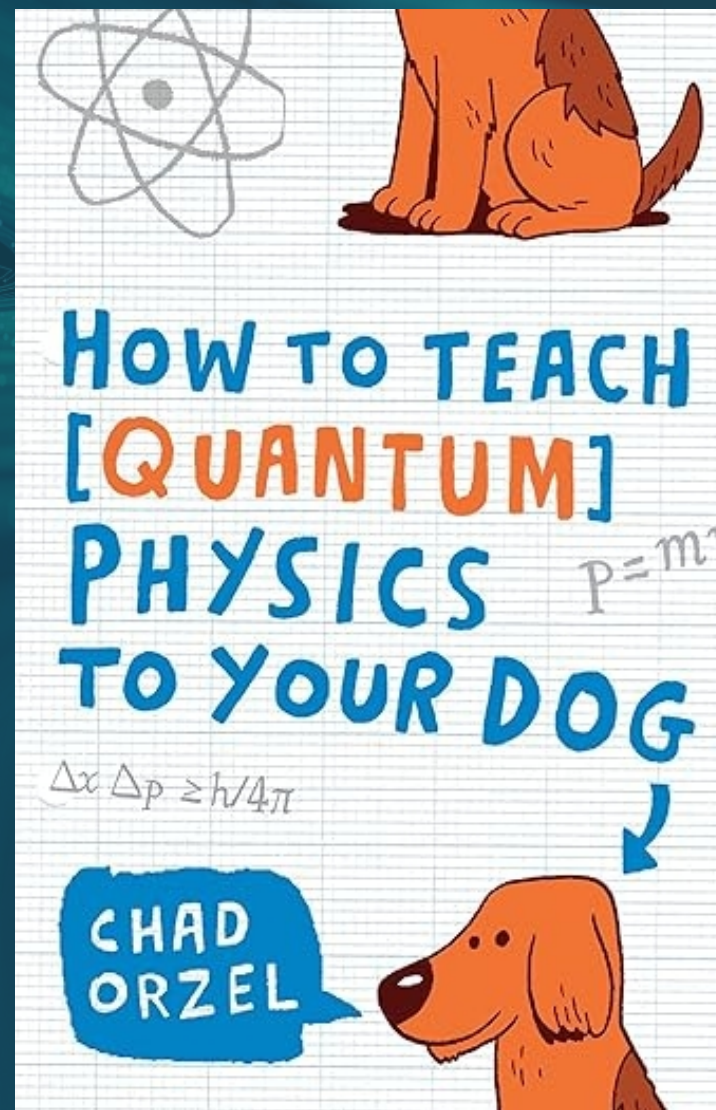
03.

穿墙而过的最短时间

02

如何与你的狗聊量子物理

With great humor and clarity, Chad Orzel explains to Emmy, and to human readers, just what quantum mechanics is and how it works—and why, although you can't use it to catch squirrels or eat steak, it's still bizarre, amazing, and important to every dog and human.





Matter Waves

感谢观看

THANK YOU FOR WATCHING

Zephyr

